Team 2 - AI Usage Journal

ARI2129: Principles of Computer Vision for AI

Topic: Histogram Equalisation and CLAHE

L-Università ta' Malta
Faculty of ICT, Department of Artificial Intelligence
Academic Year 2025-2026

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Date: May 2026

Malcolm Debattista

ARI2129 | Team 2 | May 2026

Section 1 - Tools Declared

Below are the AI tools I used during this project, with a brief note on what each was for.

Tool	Purpose
Claude (Anthropic, claude-sonnet-4-6)	Used to help structure and write the Jupyter notebook narrative, and to draft and revise written content across the project.
NotebookLM (Google, 2025)	Used for research and discovery. The five further-reading papers were uploaded and queried to map how they connected and to find a teaching sequence for the topic.

Section 2 - Notable Prompts

The prompts below are the ones that most directly shaped my work. I have focused on moments that changed something, not routine interactions.

Prompt 1 - Summarising the literature (NotebookLM)

> *"give me a brief small paragraph about all these sources, what they are about, what they explain and such..."*

NotebookLM tied the five papers together in a single paragraph. It covered CLAHE applications in face recognition, spine X-ray segmentation, retinal imaging, and hardware implementations through CUDA and VLSI, and it flagged CLALHE as an automated extension of standard CLAHE that removes manual parameter tuning. That gave me a working map of the literature before I finished reading all five papers, which saved time but also meant I came in with assumptions I later had to correct.

Prompt 2 - Structuring the teaching approach (NotebookLM)

> *"What is the best way for me to learn, and then teach this subject, according to information about the sources?"*

This was the prompt that changed how I built the notebook. NotebookLM returned a four-phase sequence grounded in the papers: start with pixel-level operations, move through global histogram equalisation, introduce CLAHE as the fix for AHE's noise problem, then finish with hardware implementation. It also suggested concept maps and pseudocode for the tiling and interpolation steps. That sequence became the section order for both the notebook and the study notes. Without it, reading five papers and working out a teaching order would have taken most of an afternoon.

Prompt 3 - Why the CDF produces a uniform output (Claude)

> *"Explain why applying the CDF as a pixel transformation function produces an approximately uniform output histogram. Keep it under 150 words and write it for second-year students."*

Claude gave a correct explanation grounded in probability: if r has CDF F(r), then s = F(r) follows a uniform distribution on [0, 1]. The logic was sound, but the notation was still too dense for the study notes audience. I kept the core argument and rewrote the framing to remove the probability symbols, replacing them with plain language about what the running total is actually doing.

Prompt 4 - Explaining the clip limit parameter (Claude)

> *"Write a Jupyter notebook markdown cell explaining what the clip limit does in CLAHE and why it prevents noise amplification. Make it conversational, not academic."*

The first draft used the phrase "robust against noise" twice and had a detached tone that did not match the rest of the notebook. I asked Claude to revise it. The second version was shorter but now described what the clip limit does without explaining why clipping at that point prevents noise. A third prompt got closer but still felt off. I ended up rewriting the cell myself. The version in the notebook is my own phrasing with one sentence kept from Claude's second attempt.

Prompt 5 - What CLALHE does differently (NotebookLM)

> *"What does CLALHE do differently from standard CLAHE, and how does it automate parameter selection?"*

NotebookLM said CLALHE selects optimal parameters automatically, without user input. That sounded straightforward, so I used it in an early draft of the study notes. When I read the actual paper, I found the automated selection still requires setting initial search bounds for the clip limit and tile size. The summary had dropped a constraint that changes how the algorithm actually works. I corrected the study notes before submission and added a sentence clarifying that CLALHE reduces manual tuning rather than removing it.

Section 3 - Five Questions

Q1. Which concept did you find hardest to explain to someone else, and why?

The hardest part was explaining why the CDF transformation produces a flat output histogram. Most people follow the steps well enough: build a histogram, compute the running total, scale to 255. But the reason those steps work sits in a probability result that is not intuitive at all. If the input intensity has CDF $F(r)$, then applying $s = F(r)$ uniformly distributes the output. Getting from "running total" to "that is why the image looks better" was where every explanation I tried fell apart. People would seem to follow it, then ask the same question a different way two minutes later.

Q2. Where did AI give you something that looked correct but turned out to be wrong or incomplete?

NotebookLM described CLALHE as automatically selecting optimal parameters without any user input. I took that at face value and put it in an early draft of the study notes. When I actually read the paper, the automated selection still requires the user to set initial search bounds for both the clip limit and the tile size. That is not a minor detail. It changes what the algorithm does and who can realistically use it. I fixed the study notes and added a sentence making the distinction clear, but I had already spent time writing around the wrong description first.

Q3. Describe a specific moment where using AI wasted your time.

I asked Claude to write a notebook markdown cell explaining the clip limit in CLAHE. The first version repeated "robust against noise" twice and read like a documentation page rather than a tutorial. I asked it to simplify. The second version was shorter but now explained what the clip limit does without touching on why it works. A third attempt got closer but the tone was still off. That exchange took about 40 minutes. I then wrote the cell myself in about ten minutes. The lesson I took from it was to just write short explanatory cells directly rather than spending three rounds trying to get an AI to write them for me.

Q4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone.

The best example is Prompt 2. I uploaded the five papers to NotebookLM and asked it how to structure the teaching of histogram equalisation based on what the sources actually said. It came back with a four-phase sequence that was grounded in the papers rather than a generic outline. That became the backbone of the notebook and informed the section order in the study notes. Working through each paper separately to build that same structure would have taken most of an afternoon. Instead I had a usable draft order in about ten minutes and spent the rest of the time building content rather than planning it.

Q5. What question about your topic does your team still not feel confident answering?

The bilinear interpolation at tile boundaries in CLAHE. The algorithm splits the image into tiles, equalises each one locally, and then interpolates between neighbouring tile mappings to avoid visible seams at the edges. Every source we read explains this at a conceptual level, but none of them work through a numerical example of the interpolation step itself. We understand what it is supposed to do. If someone asked us to implement it from scratch and show the maths at a specific boundary pixel, we would not be confident we had it right.

Julia Portelli (15506H)

ARI2129 | Team 2 | May 2026

Section 1 - Tools Declared

Below are the AI tools I used during this project, with a brief note on what each was for.

Tool	Purpose
ChatGPT	Used for explaining technical concepts, helping structure technical explanations and assessment material, improving explanations in the study notes, debugging OpenCV code, and brainstorming presentation ideas.
Claude	Used mainly for refining quiz question wording, generating alternative distractors for MCQs, and helping simplify technical explanations related to CLAHE and histogram equalisation.

Section 2 - Notable Prompts

The prompts below are the ones that most directly shaped my work on the project.

Prompt 1 - Designing adversarial quiz questions (ChatGPT)

> *"Generate difficult multiple-choice questions about histogram equalisation and CLAHE where the wrong answers are based on realistic misconceptions students commonly have."*

The AI generated quiz ideas covering histogram equalisation mappings, CLAHE clip limits, interpolation between tiles, and effects on machine learning models. Some distractors were too obvious, so I rewrote many of them manually to make the quiz more challenging and aligned with the lecture material. Working through the weak distractors helped me understand what kinds of mistakes students actually make.

Prompt 2 - Explaining the difference between HE and CLAHE (ChatGPT)

> *"Explain the difference between global histogram equalisation and CLAHE in a way that is simple enough for beginner students but still technically accurate."*

The AI explained that histogram equalisation works globally across the whole image while CLAHE processes local regions independently, and covered how clip limits reduce noise amplification. This helped improve parts of the study notes and gave a starting point for how to frame the concepts during the presentation.

Prompt 3 - Understanding CLAHE parameters (ChatGPT)

> *"Why does increasing the clip limit in CLAHE sometimes make images look noisy or artificially textured?"*

The AI explained that high clip limits allow stronger local contrast enhancement, which can amplify sensor noise in smooth regions. It also noted that smaller tiles make CLAHE more sensitive to local variation. This was useful when writing quiz questions about failure cases and parameter sensitivity.

Prompt 4 - Visual slide ideas (ChatGPT)

> *"Suggest visual slide ideas for teaching histogram equalisation and CLAHE without relying on large amounts of text."*

The AI suggested side-by-side image comparisons, histograms before and after equalisation, and diagrams showing how CLAHE divides images into tiles. These ideas fed directly into the structure and layout of the presentation slides.

Prompt 5 - OpenCV walkthrough structure (ChatGPT)

> *"Create a simple OpenCV pipeline showing histogram equalisation and CLAHE with intermediate outputs and parameter comparisons."*

The AI generated a basic pipeline using OpenCV functions for histogram equalisation and CLAHE. Some explanations were too simplified and needed expanding manually, but the structure gave a useful starting point for the notebook and suggested which intermediate visualisations to include.

Section 3 - Five Questions

Q1. Which concept did you find hardest to explain to someone else, and why?

The hardest concept was explaining how CLAHE improves local contrast while also potentially amplifying noise. Most people understand histogram equalisation as simply making an image clearer, but CLAHE behaves differently depending on the tile size and clip limit. Explaining why certain parameter choices improve detail in some images while making others look noisy or artificially textured was difficult to convey in words. Visual examples worked much better than equations for getting this across.

Q2. Where did AI give you something that looked correct but turned out to be wrong or incomplete?

During the project, AI sometimes generated explanations that looked correct on first read but turned out to be oversimplified after checking the lecture material and testing the outputs. This happened with the behaviour of CLAHE parameters, noise amplification, and practical limitations of histogram equalisation methods. Some suggested references also had incomplete citation details. To deal with it, we verified information using lecture slides, OpenCV experiments, and the research papers, then refined the explanations ourselves based on what we actually observed.

Q3. Describe a specific moment where using AI wasted your time.

AI wasted time when it generated technical summaries that sounded credible but were too generic for our project. Some responses repeated textbook definitions without covering practical limitations, failure cases, or parameter sensitivity. We tried using these in the notes and walkthrough, but they needed significant rewriting to connect properly to our examples and OpenCV outputs. The lesson was that AI is useful for generating starting points quickly, but technical academic work still requires manual verification and rewriting to be specific enough.

Q4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone.

AI helped most during the brainstorming and early development stages. It sped up generating simplified explanations, comparing ways to present concepts visually, and building initial OpenCV pipelines. It also helped structure the study notes and presentation by suggesting clearer ways to explain difficult concepts through examples. The main advantage was the speed of iteration. We could test ideas and refine explanations quickly, which meant more time improving the final quality of the work rather than starting from scratch.

Q5. What question about your topic does your team still not feel confident answering?

We are still not confident about how to choose optimal CLAHE parameters automatically for different image types. Medical images, low-light surveillance footage, and natural photographs often need different clip limits and tile sizes. We understand the general behaviour of the parameters, but we cannot define reliable rules for selecting them across image types. Testing showed that the right choices depend heavily on the specific application and image characteristics, and we do not have a principled method for making that decision.

Jake Vella (10806H)

ARI2129 | Team 2 | May 2026

Section 1 - Tools Declared

Below are the AI tools I used during this project, with a brief note on what each was for.

Tool	Purpose
ChatGPT	Used primarily for the study notes — explaining concepts, generating worked examples, and breaking down formulas in a beginner-friendly way.
Claude (Anthropic)	Used for structuring and refining the presentation slides, reducing complex explanations and suggesting visuals.

Section 2 - Notable Prompts

The prompts below are the ones that most directly shaped my work on the project.

Prompt 1 - Point processing and histogram equalisation (ChatGPT)

> *"Explain point processing in simple terms and how it relates to histogram equalisation."*

ChatGPT explained that point processing changes each pixel independently using a transformation function, and that histogram equalisation is a point processing technique because each pixel is remapped individually after the lookup table is created. This clarified the relationship between the two concepts and gave me a foundation for the study notes.

Prompt 2 - The CDF in plain language (ChatGPT)

> *"Can you explain the cumulative distribution function (CDF) in a beginner friendly way?"*

The AI described the CDF as the running total of histogram probabilities and framed it as answering the question: "How much of the image is this bright or darker?" That framing made the concept click for me and directly shaped how I explained it in the notes.

Prompt 3 - Worked numerical example (ChatGPT)

> *"Show me a worked numerical example of histogram equalisation step by step."*

ChatGPT generated a small example walking through: computing the histogram, converting counts to probabilities, calculating the CDF, building the lookup table, and mapping old values to new ones. Having a full worked example helped me understand the process mathematically rather than just memorising the steps.

Prompt 4 - Structuring the presentation (Claude)

> *"Can you help me structure my presentation accordingly."*

Claude helped reduce overly complex explanations, produced beginner-friendly summaries, and suggested visuals including histograms, CDF graphs, and before-and-after image comparisons. The main value was getting a clear presentation structure quickly rather than trying to build one from scratch.

Section 3 - Five Questions

Q1. Which concept did you find hardest to explain to someone else, and why?

The hardest concept was the CDF. I understood the mathematical steps, but explaining why accumulating probabilities actually improves contrast was difficult to put into words. It felt like just another formula rather than a meaningful transformation. After working through AI explanations and numerical examples, I understood that the CDF redistributes brightness values across a wider range, which makes image detail easier to see. That visual interpretation made it possible to explain to someone else.

Q2. Where did AI give you something that looked correct but turned out to be wrong or incomplete?

When I uploaded PDFs for the research papers, I accidentally included a file that had nothing to do with the topic. ChatGPT used it anyway and started referencing content from it without flagging anything. I noticed something had shifted in the response and asked where that information came from, which led me to find the misplaced file. The lesson was that AI does not check whether your sources are relevant — it just uses whatever you give it.

Q3. Describe a specific moment where using AI wasted your time.

When I used Claude to help with the presentation slides, it generated overly complicated examples that were not suitable for the audience. I spent extra time afterwards simplifying and restructuring almost everything it produced. The lesson was that prompts need to specify the audience and purpose upfront. Without that context, Claude defaults to a level of complexity that does not match what you actually need.

Q4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone.

AI helped most when creating worked numerical examples for histogram equalisation. It generated step-by-step calculations for histogram probabilities, CDF values, and lookup table mappings quickly. Doing that manually for multiple examples would have taken much longer. It also simplified dense paper explanations into language suitable for the presentation, which saved time when writing the script.

Q5. What question about your topic does your team still not feel confident answering?

We are still not confident about how to choose optimal CLAHE parameters for different real-world applications. We understand what the clip limit and tile size control, but there is no universal rule for selecting the right values across different image types. Medical images, facial recognition datasets, and low-light photography each behave differently, and the papers explain the effects of parameters without giving a principled method for choosing them.

Jiri Spicak

ARI2129 | Team 2 | 18 May 2026

Section 1 - Tools Declared

Below are the AI tools I used during this project, with a brief note on what each was for.

Tool	Purpose
Claude (Anthropic, Opus 4.6)	Used for generating the PowerPoint slides via pptxgenjs, extracting and explaining CLAHE concepts from research papers, and drafting this journal.
Claude (Anthropic, Opus 4.7)	Used for building the interactive CLAHE simulator.
NotebookLM (Google)	Used to organise and summarise the five source papers and generate the slide outline plan.
OpenCV documentation	Referenced for default CLAHE parameter values (clipLimit = 2.0, tileGridSize = 8x8).

Section 2 - Notable Prompts

The prompts below are the ones that most directly shaped my work on slides 16-20 and the simulator.

Prompt 1 - Feeding source literature for content extraction (Claude)

> *"Here are 5 papers about CLAHE... Can you use the ones that are actually relevant to fill in slides 16-20 from the outline I uploaded? Just ignore any papers that don't really fit the CLAHE deep-dive part."*

Claude initially tried to incorporate all five papers and build a full 20-slide deck, pulling in content from a parallel computing paper that had no relevance to my section. I had to explicitly tell it to use only those five slides and only the relevant sources. After that it correctly focused on the VLSI paper for the clip limit formula, the face recognition paper for application statistics, and the CLALHE paper for the comparison structure. The lesson was that providing too much context without explicit filtering leads to scope creep.

Prompt 2 - Generating the slide deck from the outline (Claude)

> *"Make me slides 16-20 as a .pptx file and follow the outline I uploaded exactly..."*

Claude generated a complete pptx using pptxgenjs with colour-coded cards, a 5-step process flow, tile-size comparison cards, a comparison table, icon-based application cards, and a dark-background recap slide. The first version was visually clean but content-thin — each slide had only surface-level text that would not survive follow-up questions. This led to prompt 3.

Prompt 3 - Request for more detail (Claude)

> *"I'm not that skilled in this field so I need more information in it."*

Claude rebuilt all five slides with significantly more content: concrete numbers, the clip limit formula from the source papers, OpenCV defaults, and application statistics such as 95% face recognition accuracy at 98 lux. This was the most impactful prompt because it turned surface-level slides into ones I could actually present and defend under questioning.

Prompt 4 - Building the interactive CLAHE simulator (Claude Opus 4.7)

> *"Build me an interactive simulator where you can upload an image and adjust the clipLimit using a slider from 1.0 to 10.0 and change the tileGridSize from 4x4 to 16x16, with the CLAHE output updating in real time next to the original image, and histograms of both shown side by side."*

Claude generated a working Streamlit app with dual image display, interactive sliders for both parameters, and live histogram plots. Dragging clipLimit from 2.0 to 8.0 on an X-ray visibly amplified noise in flat regions, which directly demonstrated the concept from slide 16. The simulator was the most useful artefact for understanding the material because it turned abstract theory into something I could experiment with directly.

Prompt 5 - Cross-referencing papers for the comparison table (Claude)

> *"Can you look through the VLSI paper and the CLALHE variant paper and pull out every actual difference between global HE and CLAHE... Then put it all into a comparison table for slide 18."*

Claude cross-referenced both papers and produced a structured comparison covering scope, noise handling, parameters, border artefacts, and optimal use cases. It also flagged that CLALHE introduces adaptive parameter tuning as a potential follow-up question. This was critical because it forced a structured extraction from dense academic text that I could not have done efficiently on my own.

Section 3 - Five Questions

Q1. Which concept did you find hardest to explain to someone else, and why?

The clip limit formula was the hardest. The basic idea makes sense — you cap histogram bin heights before working out the CDF. But explaining why that actually limits the maximum contrast amplification, meaning the slope of the mapping function, is where things fell apart. You have to understand the whole chain from histogram to CDF to lookup table, and each step depends on the one before it. The formula with the alpha and s_max notation made it worse because it felt abstract and I did not have a concrete image example to connect it to when explaining it out loud.

Q2. Where did AI give you something that looked correct but turned out to be wrong or incomplete?

When I uploaded the five source papers, Claude tried to build all 20 slides using every paper, including content from a parallel computing paper that was completely irrelevant to my section. I had to explicitly constrain it twice before it stayed within scope. The lesson was that giving AI too much context without clear filtering criteria produces output that looks thorough but includes material you never wanted.

Q3. Describe a specific moment where using AI wasted your time.

The first version of the slides was too thin — each slide had one or two sentences that looked fine visually but would have left me unable to answer any follow-up questions during the presentation. I had to ask Claude to rebuild everything with more depth, which meant the entire slide generation process ran twice. About 10 minutes of waiting plus reviewing slides I did not use. The lesson was to specify the required depth upfront rather than assuming the AI would match my level of comfort with the material.

Q4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone.

When I asked Claude to explain what the presentation section was actually about, it gave a clear layered explanation in about 30 seconds that would have taken me much longer to piece together from five research papers. It broke CLAHE into four steps and connected each one to why the basic approach fails. That explanation became my mental model for the entire section and shaped how I reviewed every slide. Without it I would have been presenting material I did not properly understand.

Q5. What question about your topic does your team still not feel confident answering?

The maths behind bilinear interpolation at tile boundaries. The general idea is clear — you blend nearby tile mappings so there are no hard edges. But the actual weighted combination formula, where you split the image into corner, border, and inner regions and apply different interpolation for each, is something I cannot derive from memory. The slides say smooth transitions and no tile borders, which describes the outcome without explaining the mechanics. If someone asked me to work through the distance-based weighting on a whiteboard I would struggle.